

Problem B. Starving Platform

TimeLimit: 1 second
MemoryLimit: 1024 megabytes

The Pit is a vertical prison with n floors, numbered 1 to n from top to bottom, and one prisoner on each floor. The prisoners are never let out. They are fed in a peculiar way.

Every day, a platform loaded with k portions of food starts at floor 1 and slowly descends to floor n , stopping at every floor. The prisoner on a floor may eat from the platform only while it is at their floor, and nobody saves food for the others: everyone eats until their stomach is full or the platform is empty. Whatever is left goes down to the next floor.

The stomach of prisoner i holds at most c_i portions, and it is empty on the first day. Food is never digested: once eaten, it stays in the stomach forever.

A prisoner whose stomach is still empty after the platform has passed goes hungry that day. Prisoner i dies after going hungry on h_i days in a row. Food passes the floor of a dead prisoner untouched.

More precisely, let s_i be the number of portions in the stomach of prisoner i (initially $s_i = 0$). Every day:

1. The platform visits floors $1, 2, \dots, n$ in order. If it arrives at a living prisoner i with r portions, the prisoner eats $\min(c_i - s_i, r)$ of them.
2. Every living prisoner with $s_i = 0$ goes hungry that day. A prisoner who has gone hungry on h_i consecutive days dies.

After enough days, nobody dies anymore. How many prisoners survive forever?

Input

The first line contains two integers n and k ($1 \leq n \leq 2 \cdot 10^5$, $1 \leq k \leq 10^9$): the number of floors and the number of portions on the platform each day.

The i -th of the next n lines describes the prisoner on floor i with two integers c_i and h_i ($1 \leq c_i, h_i \leq 10^9$): the size of their stomach, and how many days in a row they can go hungry before dying.

Output

Print one integer: the number of prisoners who never die.

Examples

standard input	standard output
4 3 5 2 1 2 2 3 4 2	3
5 3 4 1 2 2 3 1 1 4 5 10	4

Note

In the first example, the platform carries 3 portions every day.

- Day 1: prisoner 1 eats all 3 portions. Prisoners 2, 3 and 4 go hungry.
- Day 2: prisoner 1 eats 2 portions, which fills their stomach, and prisoner 2 eats the last portion. Prisoners 3 and 4 go hungry again; prisoner 4 has now gone hungry on $h_4 = 2$ days in a row and dies.
- Day 3: prisoners 1 and 2 are full, so prisoner 3 eats 2 portions and the last portion passes the dead prisoner 4.

From then on, prisoners 1, 2 and 3 always have food in their stomachs, so 3 prisoners survive.